

You CAN Divide by Zero! (Part 2)

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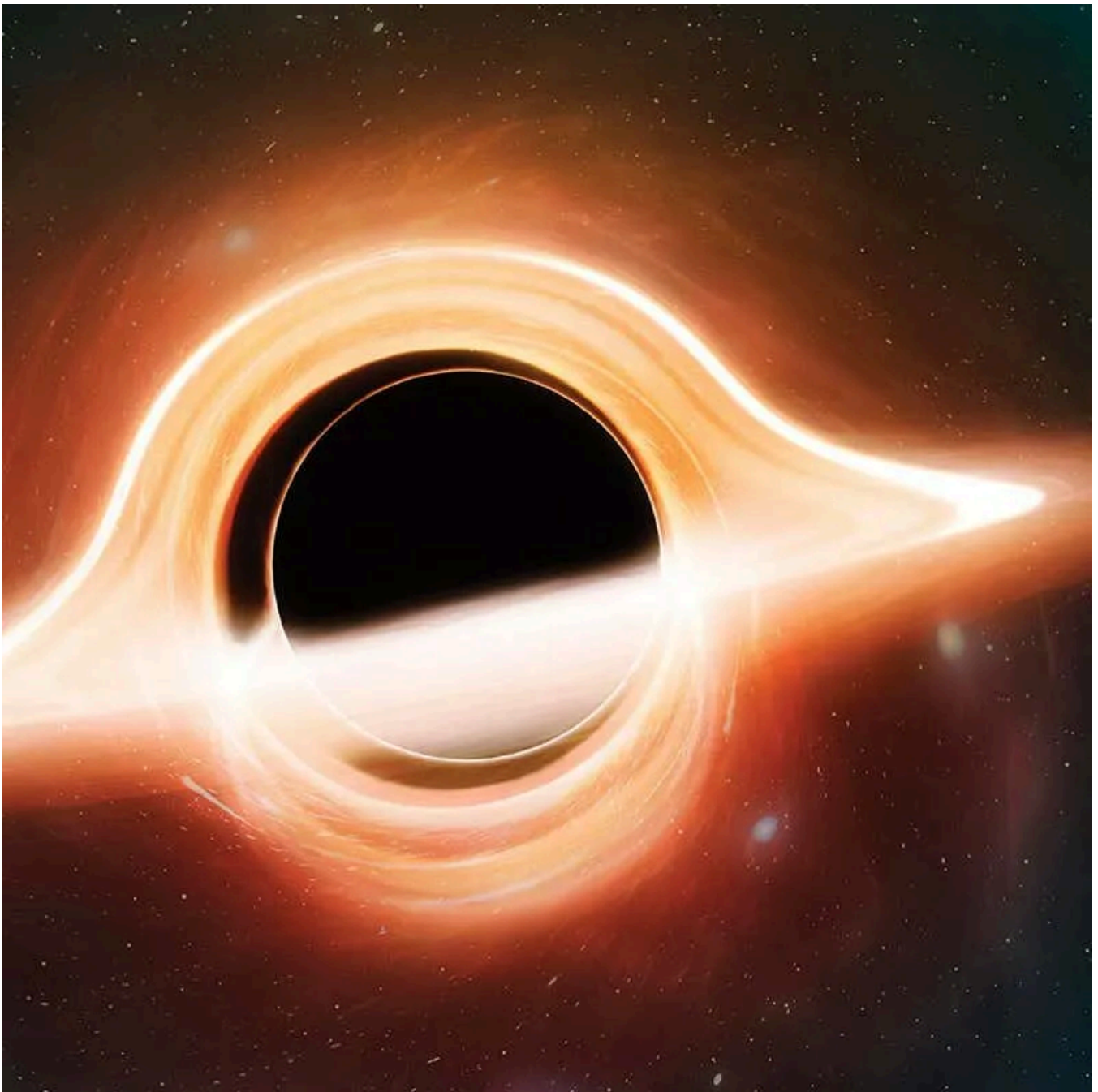
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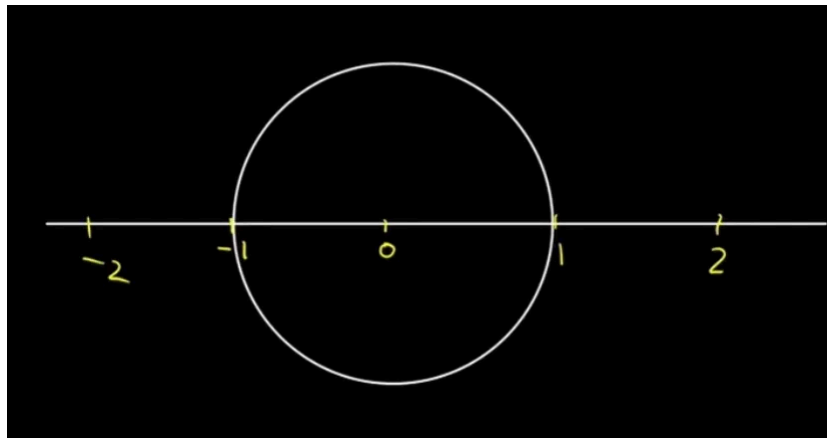


Division by 0 happens in the center of theoretical black holes

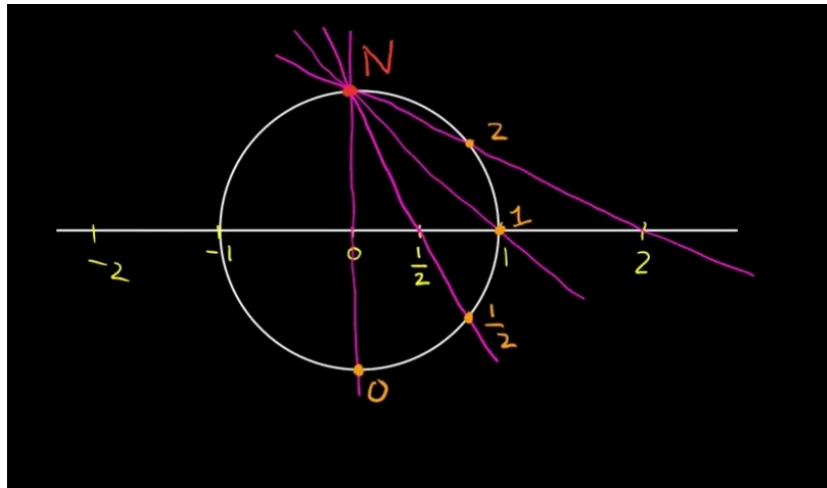
In my last article on division by zero, we talked about how it's possible to define $0/0$, albeit it will have unusual properties. In today's article, we will take this idea further and look at how we can divide other numbers by zero, including complex numbers. Let's begin.

Projective Real Line

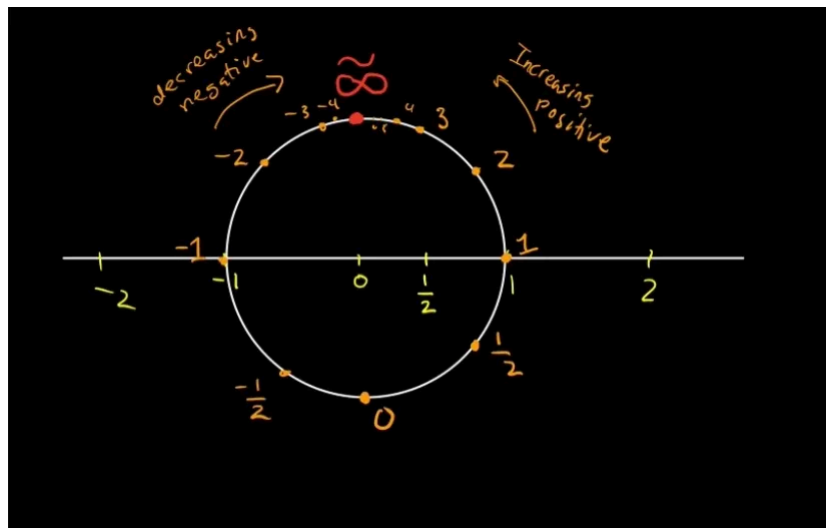
Imagine a number line extending all the way from $-\infty$ to ∞ . Now imagine a unit circle put on that number line centered around the point 1—Wait, you don't have to imagine. Here's a picture:



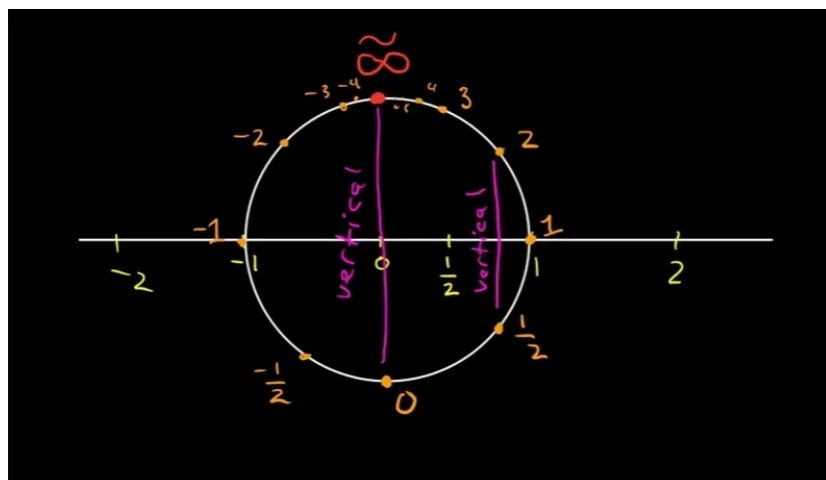
Now imagine we draw a line from each number on the number line and connect it to the North Pole of the circle. Each number will intersect the circle at exactly one point.



For example, 2 intersects the circle at the orange point labeled with “2”. We say that the orange point 2 is a *projection* of 2 onto the number line. We can see that if you project larger and larger points on the number line to the circle, they get closer and closer to the North Pole, but they never actually touch it. If we project smaller and smaller points from the negative numbers on the number line, the projections *also* get closer and closer to the North Pole. In fact, if we would project $+\infty$ from the number line to the circle, and project $-\infty$ to the circle, they would overlap at the same point. Think of taking the infinite number line and wrapping it in on itself. The ends of $-\infty$ and ∞ will conjoin to the same point. This point can’t be said to be positive infinity, nor could it be said to be negative infinity; instead, it is an *unsigned infinity*: neither positive or negative. We mark this “unsigned” infinity with a tilde.



Now of course, $+\infty$ and $-\infty$ aren't real numbers on the number line in the first place, so our above argument isn't completely rigorous. There would be a "hole" at the North Pole not having anything to be the projection of. We can then define this hole to be the unsigned infinity in order to give it meaning. Something that may have caught your eye is that 2 and $1/2$ are vertically away from each other. $1/2$ and 2 are also reciprocals of one another. It turns out that under this projection, all reciprocals on the circle will be vertically away from each other.

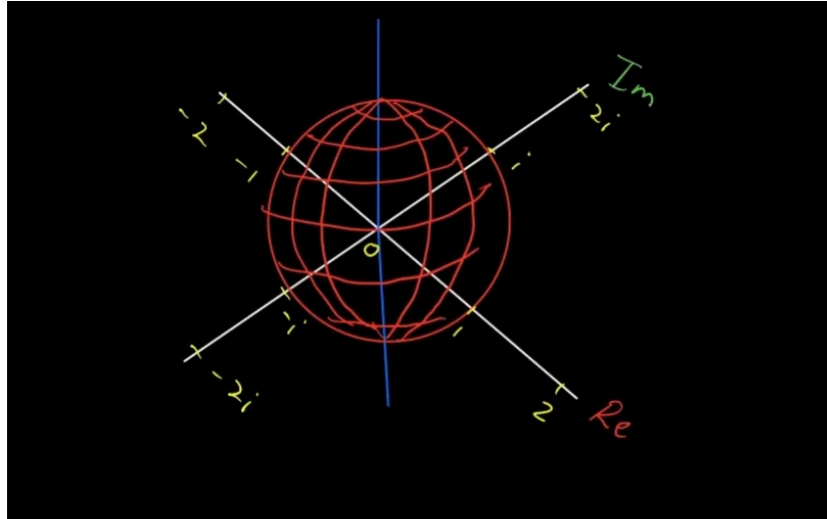


So we now have a way to divide objects on the circle. 1 divided by any number " k " will just be the number vertically away from " k ". So 1 divided by 2 is $1/2$, and 1 divided by $1/2$ is 2. It's time to address the elephant in the room staring its face right at us. 0 and unsigned infinity are *also* vertically away from each other. This means that 1 divided by 0 is unsigned infinity, and 1 divided by unsigned infinity is 0! How amazing is that! We came up with a new definition of division that satisfies what we'd like to capture, and also allows division by 0. Since $1/0 = \text{unsigned infinity}$, we can multiply by 2 on both sides to get $2/0 = 2(\text{unsigned infinity}) = \text{unsigned infinity}$.

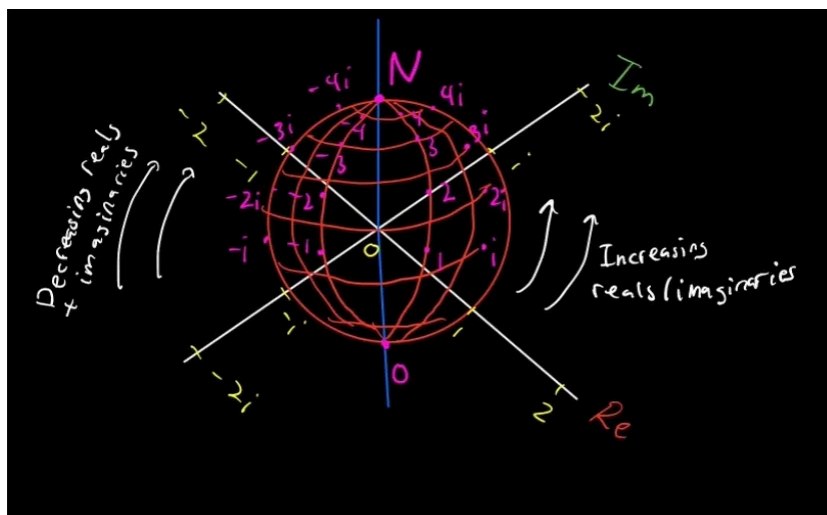
So we are allowed to divide 2 by 0 as well by defining unsigned infinity $\times$ anything = unsigned infinity.

Riemann Sphere

We don't have to restrict ourselves to the real numbers, so we can do the exact same thing on the complex plane (real line + imaginary line). The unit sphere on the complex plane looks like this:

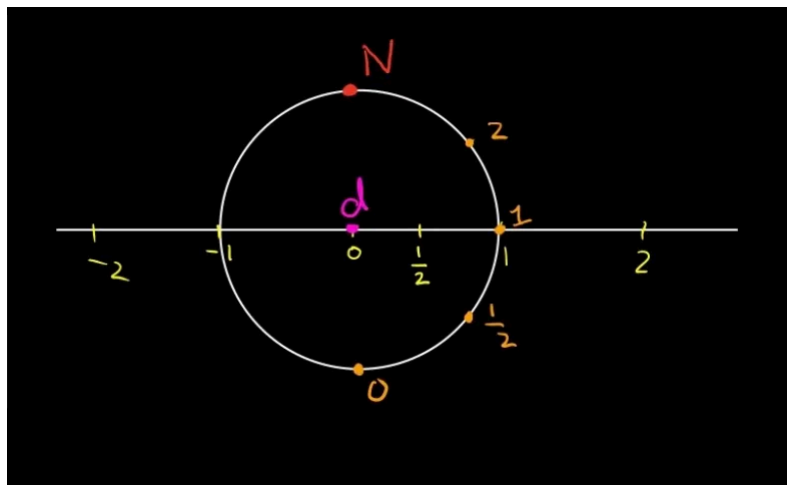


We can then project each complex number onto the sphere. Just like last time, the increasing and decreasing complex numbers approach a single point on the North Pole. This point is the **unsigned neutral infinity**. The unsigned neutral infinity is different from the unsigned infinity, since it is neither real nor imaginary nor complex. The points on the sphere “vertical” to other points are again dividing 1 by that point. This sphere has a special name by the way. It's called the **Riemann Sphere** (named after Riemann), and it has applications in complex analysis and topology. The real line projection on a circle (the one we talked about before the Riemann sphere) is called the **Projective Real Line**.



Wheel Theory

There's one subtle problem with the story we've told so far: what about unsigned infinity $\times 0$? $0 \times 1/0 = 0/0 = 0 \times$ unsigned infinity. There's no clear way to make sense of this expression on the circle, so we leave it undefined. That's a bit unsatisfactory though — have we really come all this way to *still* have $0/0$ being undefined? This is where the wheel wheels in. If you remember from the first division by zero article, we were able to divide 0 by 0 by using what is called **Wheel Theory**. We defined $0/0$ as d (called the nullity), and we defined that $d + \text{anything} = d$, and $d \times \text{anything} = d$. Any operation involving d will yield d . We can then define $0 \times$ unsigned infinity as the nullity. We can also define all the other nontrivial expressions as the nullity: $0/\text{unsigned infinity}$, $\text{unsigned infinity}/0$, $\text{unsigned infinity}/\text{unsigned infinity}$, etc.,. We can also insert the nullity onto the center of the circle, which forms the wheel shape which gives wheel theory its name. And by doing just that, we are able to consistently define division by 0 for all numbers — just don't tell your 7th grade math teacher.



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